Time-independent Schrodinger equation

Let $\Psi(x,t)$ be the wave function

$$\Psi(x,t) = \psi(x) \exp\left(-\frac{i}{\hbar} E t\right)$$

For the time derivative we have

$$i\hbar \frac{d}{dt} \Psi(x,t) = E\Psi(x,t) \tag{1}$$

Recall the Schrodinger equation

$$i\hbar \frac{d}{dt} \Psi(x,t) = H\Psi(x,t) \tag{2}$$

By equivalence of (1) and (2)

$$E\Psi(x,t) = H\Psi(x,t)$$

Let H be independent of t. Then by linearity of H the exponential can be factored out of the scope of H.

$$E\Psi(x,t) = \exp\left(-\frac{i}{\hbar} E t\right) H\psi(x)$$

Cancel exponentials to obtain the time-independent Schrodinger equation

$$E\psi(x) = H\psi(x)$$

Eigenmath script